

Problem 1. Complete the following negations:

- (i) Let $S \subset \mathbb{R}$. A function $f : S \rightarrow \mathbb{R}$ is not continuous at $c \in S$ if ...
- (ii) Let $S \subset \mathbb{R}$. A function $f : S \rightarrow \mathbb{R}$ is not uniformly continuous on S if ...
- (iii) Let $S \subset \mathbb{R}$. A sequence of functions $f_n : S \rightarrow \mathbb{R}$ does not converge uniformly to $f : S \rightarrow \mathbb{R}$ if ...

Proof. Let $S \subset \mathbb{R}$.

- (i) A function $f : S \rightarrow \mathbb{R}$ is not continuous at $c \in S$ if $\exists \epsilon > 0 \forall \delta > 0 \exists x \in S$ such that

$$|x - c| < \delta \wedge |f(x) - f(c)| \geq \epsilon.$$

- (ii) A function $f : S \rightarrow \mathbb{R}$ is not uniformly continuous on S if $\exists \epsilon > 0 \forall \delta > 0 \exists x, y \in S$ such that

$$|x - y| < \delta \wedge |f(x) - f(y)| \geq \epsilon.$$

- (iii) A sequence of functions $f_n : S \rightarrow \mathbb{R}$ does not converge uniformly to $f : S \rightarrow \mathbb{R}$ if $\exists \epsilon > 0 \forall M \in \mathbb{N} \exists n \geq M \exists x \in S$ such that

$$|f_n(x) - f(x)| \geq \epsilon.$$

□

Problem 2.

- (a) Give explicit examples satisfying the stated conditions. Explanation, but not a formal proof, for each case is required.
 - (i) A continuous function on $(0, 1)$ with neither a global maximum nor minimum.
 - (ii) A function on $[0, 1]$ with an absolute minimum at 0, absolute maximum at 1 and such that there exists $y \in (f(0), f(1))$ not in the range of f .
- (b) Let $f : S \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ be functions continuous at $c \in S$. Prove that the product $fg : S \rightarrow \mathbb{R}$ is continuous at c .

Proof. Let $f : S \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$.

- (a) We want to define f and g for each case.
 - (i) Define, for the purpose of example, $S = (0, 1)$ and $f(x) = x$. By the Pointwise Characterization of Continuity,

$$\forall x_0 \in S : \lim_{x \rightarrow x_0} f(x) = f(x_0) \implies f \in C(S, \mathbb{R}).$$

By the Perfect Completeness of $\mathbb{R}$,

$$\forall x \in S \exists \epsilon > 0 : x \pm \epsilon \in S \implies f(x - \epsilon) < f(x) < f(x + \epsilon).$$

(ii) Define, for the purpose of example, $S = [0, 1]$ and

$$g(x) = \begin{cases} x^2 & (x < 0.5) \\ \gamma & (x = 0.5) \\ x^3 & (x > 0.5) \end{cases}.$$

By the Monotone Increasing of Polynomial Functions,

$$\forall x \in S : g(0) \leq g(x) \leq g(1).$$

By the Pointwise Characterization of Discontinuity,

$$0.125, 0.250 \in (g(0), g(1)) \wedge 0.125, 0.250 \notin g(S).$$

(b) Assume f and g are continuous at $c \in S$. Suppose, for the purpose of casework, c is an isolated point of S . By the Definition of Isolated Points, $\forall \epsilon > 0 \exists \delta > 0$ such that

$$\forall x \in S \cap (c - \delta, c + \delta) : |x - c| = 0 < \delta \implies |f(x)g(x) - f(c)g(c)| = 0 < \epsilon.$$

Suppose, for the purpose of casework, c is a cluster point of S . By the Pointwise Characterization of Continuity,

$$\lim_{x \rightarrow c} f(x) = f(c) \wedge \lim_{x \rightarrow c} g(x) = g(c) \implies \lim_{x \rightarrow c} f(x)g(x) = f(c)g(c).$$

Thus, $fg : S \rightarrow \mathbb{R}$ is continuous at $c \in S$, as needed.

□

Problem 3. We say a subset $K \subset \mathbb{R}$ is compact if for every sequence $\{x_n\}$ of elements of K , there exists a subsequence $\{x_{n_k}\}$ and $x \in K$ such that $\lim_{k \rightarrow \infty} x_{n_k} = x$ (every sequence has a subsequence converging in K).

(a) Let $a < b$. Prove that $[a, b]$ is compact. *Hint:* A theorem named after two men, one with an Italian surname and the other with a German surname, might be useful.

(b) Let $a < b$, and let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Prove that

$$f([a, b]) = \{f(x) \in \mathbb{R} : x \in [a, b]\}$$

is compact.

Proof. Starto!

(a) Let $a < b$ and $\{x_n\} \subset [a, b]$. We want to prove that $\{x_n\}$ is a bounded sequence of $[a, b]$. By the Absolute Value Properties of $\mathbb{R}$,

$$\forall n \in \mathbb{N} : a \leq x_n \leq b \implies -|a| \leq x_n \leq |b| \implies |x_n| \leq |a| + |b|.$$

Then $\{x_n\}$ is a bounded sequence of $[a, b]$. We want to prove that there exists a subsequence $\{x_{n_k}\} \subset [a, b]$ and $x \in \mathbb{R}$ such that $x_{n_k} \rightarrow x$ as $k \rightarrow \infty$. By the Bolzano-Weierstrass Theorem of $\mathbb{R}$, $\exists \{x_{n_k}\} \subset [a, b]$ and $\exists x \in \mathbb{R}$ such that

$$\lim_{k \rightarrow \infty} x_{n_k} = x.$$

Then $\{x_n\}$ has a convergent subsequence in $\mathbb{R}$. We want to prove that the subsequential limit $x \in [a, b]$. By the Closure Property of $[a, b]$, $x \in \overline{[a, b]} = [a, b]$. Then $\{x_n\}$ is a closed sequence of $[a, b]$. We want to prove that $[a, b] \subset \mathbb{R}$ is compact. By the Sequential Characterization of Compactness, $[a, b] \subset \mathbb{R}$ is compact.

- (b) Let $f \in C([a, b])$ and $\{f(x_n)\} \subset f([a, b])$. We want to prove that $\{f(x_n)\}$ has a closed subsequential limit. By the Sequential Compactness of $[a, b]$, $\exists \{x_{n_k}\} \subset [a, b]$ and $\exists x \in [a, b]$ such that

$$\lim_{k \rightarrow \infty} x_{n_k} = x.$$

Then $\{x_n\}$ has a subsequential limit in $[a, b]$. By the Sequential Continuity of f ,

$$\lim_{k \rightarrow \infty} x_{n_k} = x \implies \lim_{k \rightarrow \infty} f(x_{n_k}) = f(x).$$

Then $\{f(x_n)\}$ has a subsequential limit in $f([a, b])$. We want to prove that $f([a, b]) \subset \mathbb{R}$ is compact. By the Sequential Characterization of Compactness, $f([a, b]) \subset \mathbb{R}$ is compact.

□

Problem 4.

- (a) (i) Let $a < b$. Prove that if $f : (a, b) \rightarrow \mathbb{R}$ is differentiable at $c \in (a, b)$ then f is continuous at c .
(ii) Give an explicit example of a function showing the converse of (i) is false.

(b) Let

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & (x \neq 0) \\ 0 & (x = 0) \end{cases}.$$

Use the Definition of the Derivative to prove that f is differentiable at 0.

Proof. Starto!

- (a) (i) Let $a < b$ and $f : (a, b) \rightarrow \mathbb{R}$. Assume, for the purpose of proof, f is differentiable at $c \in (a, b)$. Note that c is a cluster point of (a, b) and

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \in \mathbb{R} \implies \lim_{x \rightarrow c} \left| \frac{f(x) - f(c)}{x - c} \right| \in \mathbb{R}.$$

By the Product Relation of Limit Operations,

$$\lim_{x \rightarrow c} |f(x) - f(c)| = \lim_{x \rightarrow c} \left| \frac{f(x) - f(c)}{x - c} \right| \cdot \lim_{x \rightarrow c} |x - c| = |f'(c)| \cdot 0 = 0.$$

Note that c is not an isolated point of (a, b) and

$$\lim_{x \rightarrow c} |f(x) - f(c)| = 0 \implies \lim_{x \rightarrow c} f(x) = f(c).$$

By the Limit Characterization of Continuity, f is continuous at $c \in (a, b)$.

- (ii) Let $f : (-1, 1) \rightarrow \mathbb{R}$ and $f(x) = |x|$. Note that 0 is a cluster point of $(-1, 1)$. Compute, for the purpose of proof,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} -x = 0 = f(0)$$

and

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{|x| - 0}{x} = \lim_{x \rightarrow 0^-} -\frac{x}{x} = \lim_{x \rightarrow 0^-} -1 = -1.$$

Note that 0 is a cluster point of $(0, 1)$. Compute, for the purpose of proof,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0 = f(0)$$

and

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{|x| - 0}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1.$$

By Left-Handed and Right-Handed Characterizations of Limits,

$$\lim_{x \rightarrow 0^-} f(x) = f(0) = \lim_{x \rightarrow 0^+} f(x) \implies \lim_{x \rightarrow 0} f(x) = f(0)$$

and

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = -1 \neq 1 = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} \implies f'(0) \notin \mathbb{R}.$$

(b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & (x \neq 0) \\ 0 & (x = 0) \end{cases}.$$

Compute, for the purpose of proof, that

$$\forall x \in \mathbb{R} \setminus \{0\} : \frac{f(x) - f(0)}{x - 0} = \frac{1}{x} \left[x^2 \sin\left(\frac{1}{x}\right) - 0 \right] = x \sin\left(\frac{1}{x}\right).$$

Recall, for the purpose of proof, that

$$\forall x \in \mathbb{R} \setminus \{0\} : -1 \leq \sin\left(\frac{1}{x}\right) \leq 1 \implies -x \leq \frac{f(x) - f(0)}{x - 0} \leq x.$$

Note that 0 is a cluster point of $\mathbb{R} \setminus \{0\}$ and

$$\lim_{x \rightarrow 0} -x = 0 = \lim_{x \rightarrow 0} x.$$

By the Squeeze Theorem of $\mathbb{R}$,

$$f'(x) := \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = 0,$$

as needed.

□

Problem 5.

(a) Let $R > 0$. Prove that for all $x \in [-R, R]$,

$$e^x - \sum_{j=0}^n \frac{x^j}{j!} \leq \frac{e^R R^{n+1}}{(n+1)!}.$$

(b) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice continuously differentiable. Prove that for all $c, x \in \mathbb{R}$ with $c < x$,

$$f(x) = f(c) + f'(c)(x - c) + \int_c^x f''(t)(x - t)dt.$$

Hint: Compute the integral using a theorem which ends in parts.

Proof. Start!

(a) Let $R > 0$. We want to prove that

$$\forall x \in [-R, R] : \exp(x) - \sum_{j=0}^n \frac{x^j}{j!} \leq \frac{e^R R^{n+1}}{(n+1)!}.$$

By Pascal's Theorem, $\forall x \in [-R, R]$ such that

$$\exp(x) = e^x \implies \exp'(x) = e^x \wedge \exp^{(n)}(x) = e^x \implies \exp^{(n+1)}(x) = e^x.$$

We want to note that

$$\forall y \in (-R, R) : e^y y^{n+1} \leq e^R R^{n+1} \implies \frac{e^y y^{n+1}}{(n+1)!} \leq \frac{e^R R^{n+1}}{(n+1)!}.$$

By Taylor's Theorem, $\forall x \in [-R, R] \exists c \in (-R, R)$ such that

$$\exp(x) = \sum_{j=0}^n \frac{x^j}{j!} + \frac{e^c c^{n+1}}{(n+1)!} \implies \exp(x) - \sum_{j=0}^n \frac{x^j}{j!} \leq \frac{e^R R^{n+1}}{(n+1)!}.$$

(b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ have two continuous derivatives. We want to prove that

$$\forall c < x < \infty : f(x) = f(c) + f'(c)(x - c) + \int_c^x f''(t)(x - t)dt.$$

Assume, for the remainder of proof, $x \in (c, \infty)$. By the Integration by Parts Theorem,

$$\int_c^x f''(t)(x - t)dt = f'(t)(x - t) \Big|_{t=c}^{t=x} - \int_c^x f'(t)(x - t)'dt.$$

We want to note that

$$f'(t)(x - t) \Big|_{t=c}^{t=x} = f'(x)(x - x) - f'(c)(x - c) = -f'(c)(x - c).$$

By the Fundamental Theorem of Calculus,

$$\int_c^x f''(t)(x - t)dt = -f'(c)(x - c) + \int_c^x f'(t)dt = f(x) - f(c) - f'(c)(x - c).$$

Indeed,

$$f(x) = f(c) + f'(c)(x - c) + \int_c^x f''(x - t)dt,$$

as needed.

□

Problem 6.

(a) Compute

$$\int_0^1 x \log x dx := \lim_{a \rightarrow 0^+} \int_a^1 x \log x dx.$$

You may use without proof this version of **L'Hôpital's Rule**: if $f, g : (0, 1) \rightarrow \mathbb{R}$, $g(x) \neq 0$ for all x , $f(x) \rightarrow -\infty$, $g(x) \rightarrow \infty$ as $x \rightarrow 0^+$, and $L = \lim_{x \rightarrow 0^+} f'(x)/g'(x)$ exists, then $\lim_{x \rightarrow 0^+} f(x)/g(x) = L$.

(b) Prove

$$\lim_{n \rightarrow \infty} \int_0^1 x^n \sin(x) dx = 0.$$

(c) Let $f : [-\pi, \pi] \rightarrow \mathbb{R}$ be a continuously differentiable function. Prove that

$$\lim_{n \rightarrow \infty} \int_{-\pi}^{\pi} \sin(nx) f(x) dx = 0$$

Hint: $\sin(nx) = [-\frac{1}{n} \cos(nx)]'$.

Proof. Start!

(a) We want to compute that

$$\int_0^1 x \log x dx := \lim_{a \rightarrow 0^+} \int_a^1 x \log x dx.$$

By the Integration by Parts Formula,

$$\begin{aligned} \int_a^1 x \log x dx &= \frac{1}{2} x^2 \log x \Big|_{x=a}^{x=1} - \frac{1}{2 \ln 10} \int_a^1 x dx \\ &= \frac{1}{2} x^2 \log x - \frac{1}{4 \ln 10} x^2 \Big|_{x=a}^{x=1} = \frac{1}{4 \ln 10} (a^2 - 1) - \frac{1}{2} a^2 \log a \end{aligned}$$

We want to note that

$$\lim_{a \rightarrow 0^+} \left[\frac{1}{4 \ln 10} (1 - a^2) \right] = \frac{1}{4 \ln 10}.$$

By the L'Hôpital's Indeterminate Rule,

$$\lim_{a \rightarrow 0^+} \left[\frac{1}{2} a^2 \log a \right] = \frac{1}{2} \lim_{a \rightarrow 0^+} \frac{\log a}{a^{-2}} \left(\frac{-\infty}{+\infty} \right) = -\frac{1}{4 \ln 10} \lim_{a \rightarrow 0^+} a^2 = 0.$$

Thus,

$$\int_0^1 x \log x dx = \lim_{a \rightarrow 0^+} \left[\frac{1}{4 \ln 10} (a^2 - 1) \right] - \lim_{a \rightarrow 0^+} \left[\frac{1}{2} a^2 \log a \right] = -\frac{1}{4 \ln 10},$$

as needed.

(b) We want to prove that

$$\lim_{n \rightarrow \infty} \int_0^1 x^n \sin(x) dx = 0.$$

By the Triangle Inequality of Integration,

$$\left| \int_0^1 x^n \sin(x) dx \right| \leq \int_0^1 |x^n \sin(x)| dx = \int_0^1 x^n |\sin(x)| dx \leq \int_0^1 x^n dx.$$

We want to note that

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_{x=0}^{x=1} = \frac{1}{n+1}.$$

By the Squeeze Theorem of Limitation,

$$-\frac{1}{n+1} \leq \int_0^1 x^n \sin(x) dx \leq \frac{1}{n+1} \implies \lim_{n \rightarrow \infty} \int_0^1 x^n \sin(x) dx = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0,$$

as needed.

- (c) Let $f : [-\pi, \pi] \rightarrow \mathbb{R}$ be any continuously differentiable function. We want to prove that

$$\lim_{n \rightarrow \infty} \int_{-\pi}^{\pi} \sin(nx) f(x) dx = 0.$$

By the Integration by Parts Formula,

$$\int_{-\pi}^{\pi} \sin(nx) f(x) dx = -\frac{1}{n} \cos(nx) f(x) \Big|_{x=-\pi}^{x=\pi} + \int_{-\pi}^{\pi} \frac{1}{n} \cos(nx) f'(x) dx$$

We want to note that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \cos(nx) f(x) \Big|_{x=-\pi}^{x=\pi} = (f(\pi) - f(-\pi)) \lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0.$$

By the Squeeze Theorem of Limit Operations,

$$\left| \int_{-\pi}^{\pi} \frac{1}{n} \cos(nx) f'(x) dx \right| \leq \frac{1}{n} \int_{-\pi}^{\pi} |f'(x)| dx \implies \lim_{n \rightarrow \infty} \int_{-\pi}^{\pi} \frac{1}{n} \cos(nx) f'(x) dx = 0.$$

Thus,

$$\lim_{n \rightarrow \infty} \int_{-\pi}^{\pi} \sin(nx) f(x) dx = 0 + 0 = 0,$$

as needed. □

Problem 7.

- (a) Give an explicit example of a sequence of continuous functions on $(0, 1)$ that converges pointwise to a continuous function on $(0, 1)$ but the convergence is not uniform. Explanation, but not a formal proof, is required. Feel free to use pictures.
- (b) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function and there exists $L > 0$ such that

$$|f'(c)| \leq L \quad \forall c \in \mathbb{R}.$$

- (i) Prove that $f : \mathbb{R} \rightarrow \mathbb{R}$ is Lipschitz continuous on $\mathbb{R}$.
- (ii) Define a sequence of functions $f_n : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f_n(x) := f\left(x + \frac{1}{n}\right).$$

Prove that $\{f_n\}$ converges to f uniformly on $\mathbb{R}$.

Proof. Start!

- (a) Let $f_n(x) = \sqrt[n]{x}$ and $f(x) = 1$ for $x \in (0, 1)$. Then $f_n \in C((0, 1))$ and $f \in C((0, 1))$. We want to prove that f_n converges pointwise to f on $(0, 1)$. Compute, for the purpose of example, the pointwise limit

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \sqrt[n]{x} = 1 = f(x).$$

Then f_n converges pointwise to f on $(0, 1)$. We want to prove that f_n does not converge uniformly to f on $(0, 1)$. By the Limsup Characterization of Uniform Convergence, $f_n(x)$ does not converge uniformly to $f(x)$ on $(0, 1)$ if and only if

$$\limsup_{n \rightarrow \infty} |f_n(x) - f(x)| \neq 0.$$

We want to note that $f_n(x) \in (0, 1)$ for $x \in (0, 1)$. Compute, for the purpose of example, the supremum

$$\sup_{x \in (0,1)} |f_n(x) - f(x)| = \sup_{x \in (0,1)} |\sqrt[n]{x} - 1| = \sup_{x \in (0,1)} (1 - \sqrt[n]{x}) = 1$$

and the uniform limit

$$\limsup_{n \rightarrow \infty} |f_n(x) - f(x)| = \lim_{n \rightarrow \infty} \left(\sup_{x \in (0,1)} |f_n(x) - f(x)| \right) = \lim_{n \rightarrow \infty} 1 = 1.$$

Then f_n does not converge uniformly to f on $(0, 1)$.

- (b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function. Assume, for the remainder of proof, $\exists L > 0 \forall c \in \mathbb{R}$ such that

$$|f'(c)| \leq L.$$

- (i) By the Mean Value Theorem, $\forall x, y \in \mathbb{R} \exists c \in (x, y) \cup (y, x)$ such that

$$|f(x) - f(y)| = |f'(c)(x - y)| = |f'(c)||x - y| \leq L|x - y|.$$

Thus, f is Lipschitz continuous on $\mathbb{R}$, as needed.

- (ii) Define, for the remainder of proof, $f_n : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f_n(x) := f\left(x + \frac{1}{n}\right).$$

Let $\epsilon > 0$. Choose $M \in \mathbb{N}$ such that $M\epsilon > L$. If $x \in \mathbb{R}$ and $n \geq M$, then

$$|f_n(x) - f(x)| = \left| f\left(x + \frac{1}{n}\right) - f(x) \right| \leq L \left| x + \frac{1}{n} - x \right| = \frac{L}{n} \leq \frac{L}{M} < \epsilon.$$

Thus, f_n converges uniformly to f on $\mathbb{R}$ as needed.

□